DESKTOP UI DESIGN PROPOSAL

AEGIS BROWSER

Liquid Glass Glassbar

A zero-chrome concept for Aegis on Linux & Windows: every page runs edge to edge in its own window, while tabs, the omnibox and the menu live inside one liquid-glass capsule — and collapse to a six-pixel rail on the left edge.

Concept v1.0 · Interaction & Visual Specification

Hardened by default. Quiet by design.

AEGIS-BROWSER.PAGES.DEV · NOVEMBER 2026

Contents

01 · Design Concept — Zero Chrome, Desktop Edition 1

02 · The Glassbar — Capsule, Strip and Edge Rail 2

2.1 The strip — hover state . . . 3

2.2 The edge rail — collapsed state . . . 3

2.3 The trust line . . . 4

03 · Motion, Hover and Auto-Hide 4

3.1 Auto-hide behavior . . . 5

04 · Overlays — Tab Grid, Menu, Command Omnibox 6

05 · Design Tokens 7

06 · Platform Implementation Notes 9

6.1 Ship sequence at a glance . . . 12

01 · Design Concept — Zero Chrome, Desktop Edition

Aegis Browser is a privacy-hardened Chromium build whose identity already promises to be **hardened by default, quiet by design** — and its mobile concept (see the companion proposal) keeps that promise by folding all chrome into a liquid-glass dock. The desktop, ironically, is where browsers are loudest. A conventional desktop layout bolts a 40 px tab strip and a 48 px toolbar across the full width of every window — 9.8% of a 1,440 × 900 viewport before the bookmarks bar even enters the conversation. On a 13-inch laptop at 125% scaling the same furniture eats closer to 13% of a screen the user paid for by the diagonal inch.

This proposal extends the same answer to Linux and Windows. **The page is the window**: every site loads edge to edge in a frameless surface, and the browser retreats into a single floating **liquid-glass capsule** — the Glassbar — that carries the trust line, the tab strip, the omnibox and the menu in one 640 × 44 pill docked to the top of the window. When the user reads, the capsule thins into a **6 × 160 edge rail** on the left, the direct sibling of the mobile sliver. One material, two platforms: desktop and Android Aegis read as the same product, not two products that share a name.

2.2%
of the viewport covered by the compact capsule — versus ≈9.8% for a conventional tab strip + toolbar, a 4.5× reduction

0.07%
of the viewport covered by the collapsed edge rail — a six-pixel line the article simply flows past

1 keystroke
reaches every frequent action — omnibox, tab grid, rail toggle, private window — no pointer travel required

The visual language is unchanged from the mobile edition because it should be: strict monochrome, true blacks and paper whites, Montserrat for identity type, and a single material — liquid glass — between the user and the page. In dark theme the glass is smoked ink; in light theme it is frosted paper; both are the same five-layer recipe with different tokens. What the desktop adds is **hover** as a first-class citizen: the capsule expands to a full tab strip under the cursor and retracts when the mouse leaves, so the browser grows and shrinks around attention rather than around taps.

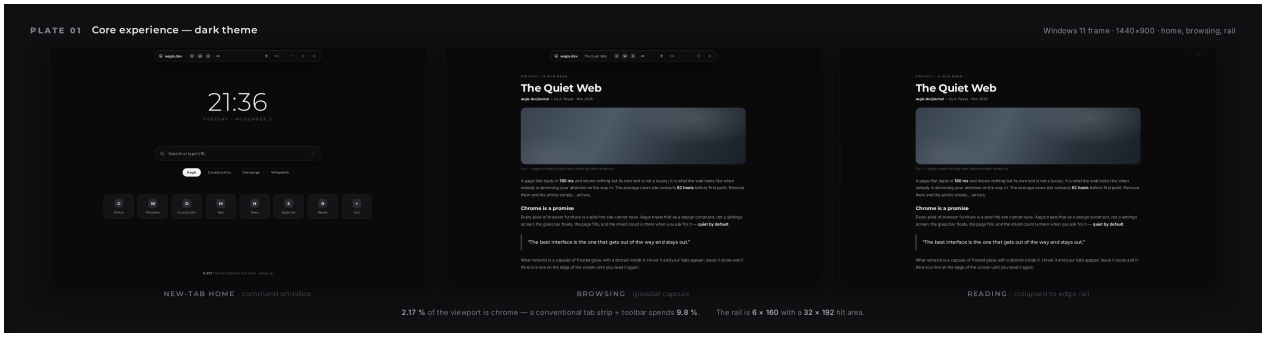


Plate 01 — Core experience, dark theme, Windows 11 frame: new-tab home · browsing with the capsule · reading with the rail.

The home screen is a command surface, not a dashboard: a clock, one search pill, engine chips and eight tiles — nothing that cannot be rendered from local data, because a hardened browser has no business phoning home to draw its own start page. While browsing, the same capsule shows the domain with the shield glyph; while reading, even that yields to the rail. Everything else — tabs, menu, omnibox — opens as glass sheets above the untouched page.

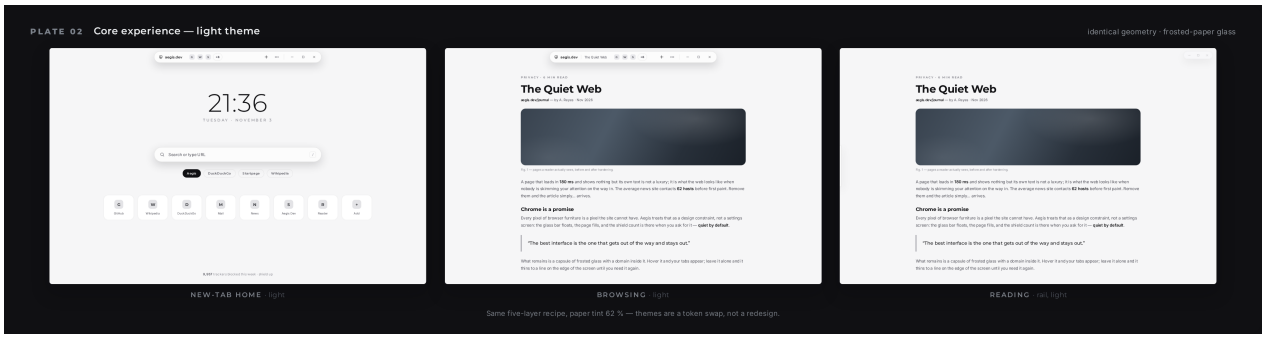


Plate 02 — Core experience, light theme: identical geometry in frosted-paper glass. Every state in this proposal is specified for both themes.

02 · The Glassbar — Capsule, Strip and Edge Rail

The Glassbar is the entire browser chrome compressed into one capsule: **640 × 44 dp**, corner radius 22, floating 12 dp below the top edge of the window. Its segments, in reading order: the **trust segment** (shield glyph, domain, current page title), up to **three favicon tabs** plus a "+N" overflow chip, the **new-tab** and **menu** buttons, and the **caption cluster** — minimize, maximize and close rendered inside the glass, because on a zero-chrome desktop the title bar has been absorbed into the browser. The full top band of the window is the drag region; double-click maximizes; dragging to screen edges triggers the platform snap layout exactly as users expect.

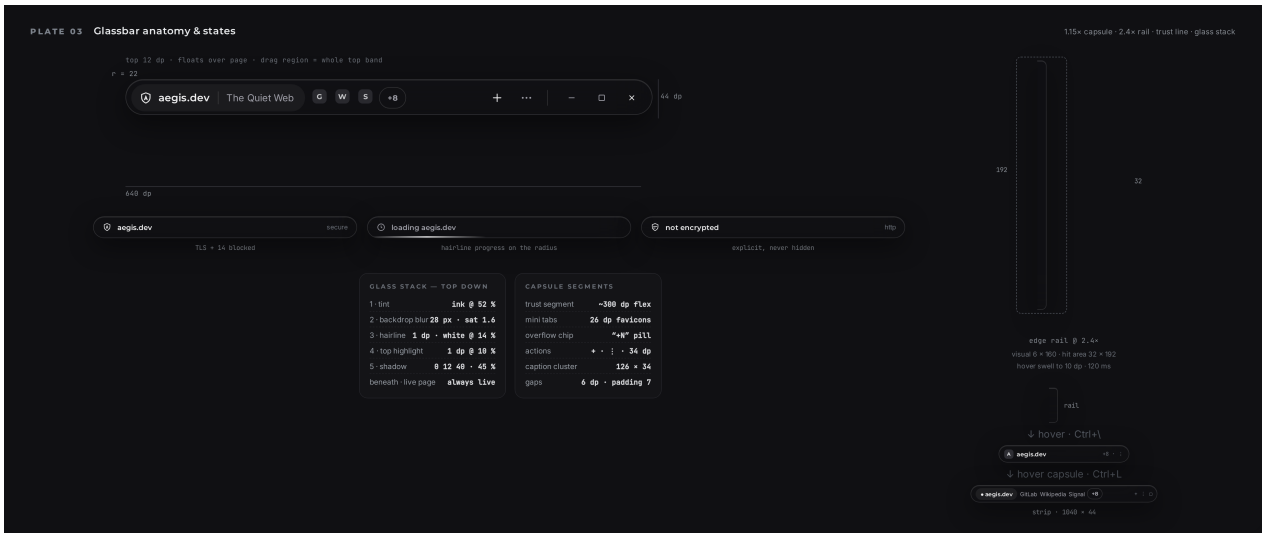


Plate 03 — Glassbar anatomy at 1.15×: capsule segments and dimensions · the three trust states · the five-layer glass stack · the rail at 2.4× with its generous hit area · the state chain rail → capsule → strip.

Table 1 — Capsule geometry (compact state)

Element	Specification	Notes
Container	640 × 44 dp, radius 22	Full capsule; floats 12 dp below top edge, centered
Trust segment	≈300 dp flex, radius 17	Shield glyph + domain + page title, truncates from the right
Mini tabs	3 × 26 dp favicons	Inactive tabs, 8 dp radius tiles; active tab lives in the trust segment
Overflow chip	"+N" pill, 11 dp type	Opens the tab grid (Section 04) — N is the hidden count
Action buttons	34 dp circles: + and ⋮	New tab; menu (Alt+E). Long-press + reopens closed tabs
Caption cluster	126 × 34 dp	Windows: minimize / maximize / close. GNOME CSD: close only
Spacing	6 dp gaps, 7 dp padding	Capsule padding; 4 dp inside the tab cluster

All values in device-independent pixels at the 1,440 × 900 baseline; scale with OS display factor.

The glass is the mobile recipe applied at desktop scale: a 28 px backdrop blur with 1.6× saturation, a monochrome tint (52% ink in dark theme, 62% paper-white in light), a 1 dp hairline border, a 1 dp inner top highlight and a soft drop shadow. No layer is opaque — the live window content remains faintly visible through the capsule, which is what makes the bar feel suspended over the page rather than painted on it.

2.1 The strip — hover state

Hovering the capsule (or pressing Ctrl+L) expands it smoothly to a **1,040 × 44 strip**: labeled tabs with close buttons replace the favicons, the overflow chip becomes "+7", and the trust segment keeps its place at the left. The expansion is a width transition — 240 ms, emphasized decelerate — not a reflow; the capsule and the strip are the same view at two widths. When the pointer leaves the top band, the strip relaxes back to the capsule after 800 ms, so passing the mouse across the top of the window on the way to a bookmark never punishes the reader with a permanent toolbar.

2.2 The edge rail — collapsed state

When the user starts reading, the capsule flows left and thins into a **6 × 160 dp rail**, flush against the left edge and vertically centered — the desktop sibling of the mobile sliver. The rail keeps the glass recipe and a 32 × 192 hit area; hovering swells it to 10 dp in 120 ms as a live affordance. A click, a short swipe right or Ctrl+\ recalls the capsule. While the rail is collapsed, a faint caption cluster fades in at the top-right corner of the window (38% opacity, full on hover) so window controls remain reachable without expanding the chrome.

Table 2 — Edge rail specification

Property	Value	Notes
Size (idle)	6 × 160 dp	Rounded 4 dp on the inner corners, flush left
Size (hover)	10 × 160 dp	Proximity swell, 120 ms ease-out
Position	left: 0, vertically centered	Inside the window, clear of snap-zone hotspots
Hit area	32 × 192 dp	Comfortable mouse target despite 6 dp visuals
Recall actions	Click; swipe right; Ctrl+\; edge hover	Capsule restores with a 360 ms spring
Corner caption	126 × 34 @ 38% → 100% on hover	Window controls remain reachable while collapsed
Fullscreen (F11)	Rail fades out after 4 s	Top-edge or left-edge hover restores it temporarily

The rail inherits the capsule recipe unchanged — one material, three scales (rail 6 dp, capsule 44 dp, strip 44 dp wide).

2.3 The trust line

As on mobile, the trust segment is the only place the browser is allowed to speak, and it speaks only about trust: shield glyph and domain on a secure connection, a hairline progress curve riding the capsule's bottom radius while loading, and an explicit **"not encrypted"** treatment for plain HTTP. Trackers blocked on the current page surface as a count inside the menu header — reassurance on demand, never ambient anxiety. Private windows invert the trust pill and the tab tiles, so a screenshot of an Aegis desktop window always tells you which mode it was taken in.

03 · Motion, Hover and Auto-Hide

Desktop motion has two inputs the phone never had — the pointer and the keyboard — and Aegis treats both as signals of intent. Every transition completes in under half a second and every curve belongs to one family: fast departure, gentle landing, cubic-bezier(.3, 0, .1, 1) for the signature collapse. The capsule never "opens" and the rail never "appears"; one material shrinks into the edge and stretches back, so the state change reads as a single continuous object at three widths.

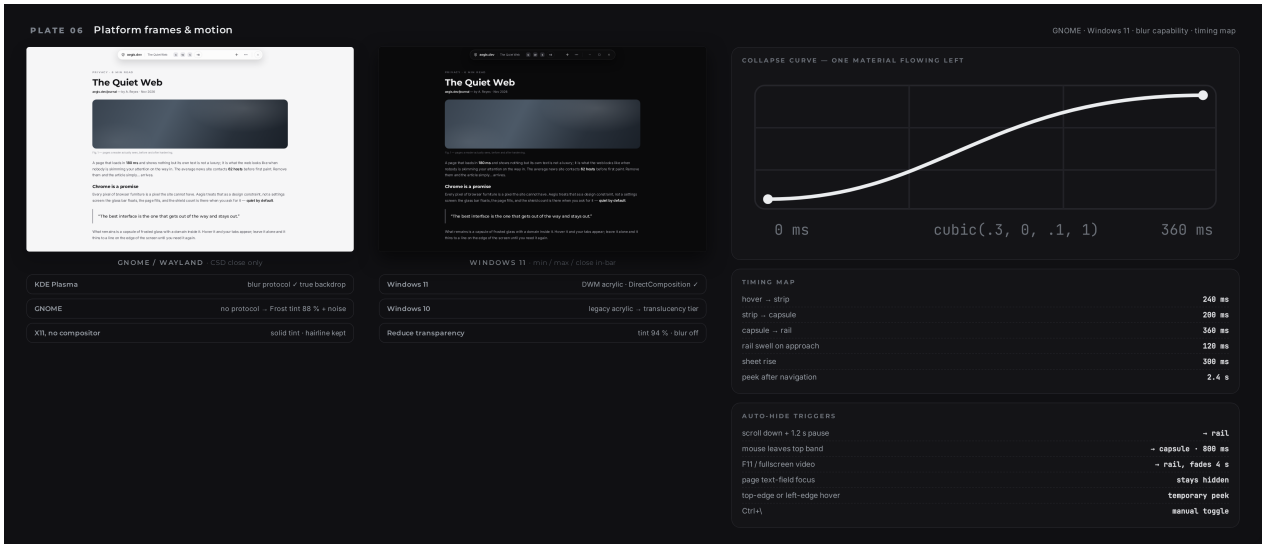


Plate 04 — Platform frames and motion: GNOME / Wayland and Windows 11 renderings · blur capability by platform · the collapse curve, timing map and auto-hide triggers.

Table 3 — Pointer and keyboard map

Input	Context	Action
Hover capsule / Ctrl+L	Capsule	Expand to strip; focus omnibox on keystroke
Pointer leaves top band	Strip	Relax to capsule after 800 ms
Click rail / swipe right / Ctrl+\	Rail	Recall the capsule (360 ms)
Scroll down + 1.2 s pause	Any chrome state	Collapse to rail
Ctrl+T · Ctrl+W · Ctrl+Shift+T	Anywhere	New tab · close tab · reopen closed
Ctrl+Tab / Ctrl+Shift+Tab	Anywhere	Cycle tabs with a 9-tile ghost preview
Ctrl+Shift+A	Anywhere	Open the tab grid overlay
Middle-click tab / title	Strip, tab grid	Close tab (reopen from menu › History)
F11 · Esc	Page · overlay	Fullscreen (rail auto-fades) · dismiss sheet

Pointer gestures bind to glassbar chrome and window edges only — page-level interactions are never intercepted.

3.1 Auto-hide behavior

The Glassbar manages itself around reading. After each navigation completes it performs a 2.4-second peek so the trust line is readable for the fresh page, then retreats. Scrolling down and pausing 1.2 seconds collapses it; the first upward scroll restores the capsule. Fullscreen video hides everything and fades the rail after four seconds — moving the mouse to the top or left edge peeks it back. Focusing a text field on the page keeps the chrome hidden so the IME owns the bottom edge, and a "reading mode" setting (default on) makes the rail permanent for users who never want to see chrome mid-article. Under the system "reduce motion" setting

every transition collapses to an 80 ms fade.

04 · Overlays — Tab Grid, Menu, Command Omnibox

Three glass sheets do the rest of the work, and all three obey the same physics: they rise from the capsule or the top edge, they blur the live page behind them, and they dismiss with Esc or a click on the page. Because the sheets are made of the same liquid glass as the capsule, the interface never introduces a second material — panels feel like the bar pulled open rather than dialogs dropped on top. The scrim is a 2 px blur plus a 46% ink wash (28% in light theme): enough to focus, never enough to hide the page.

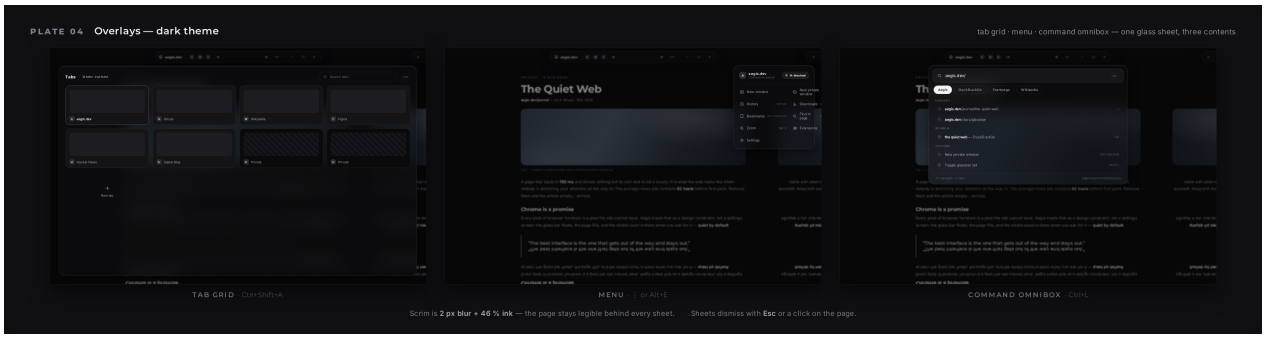


Plate 05 — Overlays, dark theme: tab grid with inverted private cards · menu with live shield count · command omnibox with grouped suggestions.

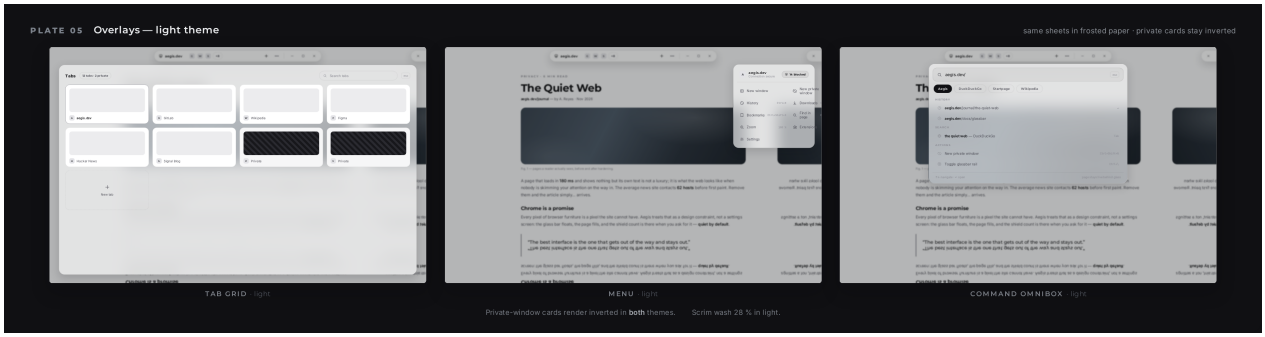


Plate 06 — Overlays, light theme: the same three sheets in frosted paper. Private cards stay inverted in both themes.

Table 4 — Overlay specification

Overlay	Entry point	Contents	Dismiss
Tab grid	Ctrl+Shift+A; "+N" chip; menu	Search filter; 4-column cards with favicon, title and live thumbnail; private cards inverted; "New tab" tile	Esc; click page; Ctrl+W closes focused card
Menu	⋮ button; Alt+E	Domain header with "N blocked" shield pill; 2-column grid: new / private window, history, downloads, bookmarks, find, zoom, extensions; Settings row	Esc; click scrim; fling up
Command omnibox	Ctrl+L; click capsule; "/" on home	Focused input with caret; engine chips (Aegis, DuckDuckGo, Startpage, Wikipedia); grouped history, search and actions with shortcut hints	Esc; Enter opens; Tab switches engine

Sheets cap at 82% of window height; longer content scrolls inside the sheet with the header pinned.

05 · Design Tokens

Tokens are the contract between this proposal and the build, shared verbatim with the mobile edition wherever the words "glass", "ink" or "paper" appear. Every value is theme-paired — dark first, light in lockstep, no orphan colors. The palette is achromatic: hierarchy is carried by lightness and blur, not hue, so the browser never competes with whatever colors the web page chooses to be. Pure white in dark theme and pure ink in light theme act as the only accents, and both themes pass AA contrast on their surfaces.

Table 5 — Core palette

Token	Dark value	Light value	Usage
bg / page	#0A0A0B	#F6F6F7	Window background behind content
surface	#141416	#FFFFFF	Cards, tab thumbnails, sheets base
text / primary	#F4F4F5	#101013	Headings, active icons, tab titles
text / secondary	#A1A1AA	#52525B	Body copy, labels
text / tertiary	#63636B	#8E8E96	Meta, captions, hints
line / hairline	white @ 10%	ink @ 8%	Dividers, card borders
accent (invert)	#FFFFFF	#0A0A0B	Primary buttons, active engine chip
scrim	ink @ 46%	ink @ 28%	Behind overlays

One grey ramp; tokens are shared with the mobile proposal, which is what makes the two platforms read as one product.

Table 6 — Liquid glass recipe

Layer	Dark theme	Light theme
Backdrop blur	28 px, saturate 1.6, brightness 1.06	28 px, saturate 1.4
Tint	rgba(16, 16, 18, 0.52)	rgba(255, 255, 255, 0.62)
Border	1 dp, white @ 14%	1 dp, ink @ 12%
Top highlight	inset 1 dp, white @ 10%	inset 1 dp, white @ 55%
Shadow	0 12 px 40 px, black @ 45%	0 12 px 34 px, ink @ 16%

Applied identically to capsule, rail, sheets and tiles — one material everywhere, on both platforms.

Table 7 — Typography (Montserrat + Inter)

Role	Spec	Usage
Display / clock	96 dp / 300	New-tab home clock, tabular figures
Omnibox input	16 dp / 400	Command omnibox and URL editing
Tab title	12 dp / 400 · active 500	Strip tabs; truncates with ellipsis
Trust pill	13 dp / 600	Domain in the capsule
Overlay title	16 dp / 700	Sheet headers ("Tabs", domain)
Body / UI	13 dp / 400 · 1.6	Menus, suggestions, labels
Micro label	10 dp / 600 · +8% tracking, caps	Group headers, engine chips, kbd hints

Montserrat carries identity and numerals; Inter carries dense UI text. JetBrains Mono for shortcut hints.

Table 8 — Shape, spacing, elevation

Token	Value	Notes
Radius / capsule & strip	22 dp (full pill)	Half of 44 dp height
Radius / rail	4 dp inner corners	Flush left edge
Radius / sheets	22 dp	All sheets; cards 16, tiles 18
Spacing unit	4 dp base · 8/12/16/24 steps	Capsule gaps 6, sheet padding 22–24
Elevation 0	Glass only	Capsule, rail, tiles
Elevation 1	Glass + shadow 40 px blur	Capsule over scrolled content
Elevation 2	Glass + shadow + scrim	All three sheets

Iconography: the same 24 dp outline set as mobile, 1.7 dp round-capped strokes, pure currentColor.

06 · Platform Implementation Notes

The concept is deliberately buildable inside a Chromium codebase with **no compositor surgery**. The capsule, rail and sheets are one layer in the existing Views hierarchy, drawn above the WebContents frame; the omnibox, TabModel and menu code paths are reused behind the new presentation exactly as planned for mobile. The window itself runs frameless: on Windows, WM_NCHITTEST routes the top band as caption area and the three caption buttons are Aegis-drawn; on Linux the same geometry ships as GTK CSD (Wayland xdg-decoration) with a single close button. Snap layouts, aero snap and workspace switching keep working because the drag band behaves like a native title bar.

The glass itself is where the platforms diverge, and the proposal prices that divergence honestly. Windows 11 offers real system blur through DWM (DirectComposition acrylic, or the Mica backdrop variant); KDE Plasma exposes a blur protocol the window can opt into; GNOME deliberately offers neither. Rather than fork the design per desktop, Aegis renders its own backdrop: the compositor samples the WebContents frame at quarter resolution, blurs it on the GPU (Skia) and caches the result, refreshing on scroll-idle with a 200 ms throttle. Where the OS offers a cheaper native blur (Win11, KDE) it is used; everywhere else the readback path produces the identical look — including GNOME, which is exactly the point.

Listing 1 — Glass recipe as CSS custom properties (prototype reference for the Views port)

```
/* Liquid glass — desktop recipe (tokens from Table 6) */

.glassbar {

  position: absolute; top: 12dp; left: 50%; translate: -50% 0;

  width: 640dp; height: 44dp; border-radius: 22dp;   /* full pill */

  background: var(--glass-tint);        /* ink 52% · paper 62% */

  backdrop-filter: blur(28px) saturate(1.6) brightness(1.06);

  border: 1px solid var(--glass-border);/* white 14% · ink 12% */

  box-shadow: 0 12px 40px var(--glass-shadow),

              inset 0 1px 0 var(--glass-highlight);

}

/* States: .strip = width 1040dp · .rail = 6x160dp at left:0 */

/* Fallback ladder and platform backends: see Tables 9-10 */
```

Table 9 — Platform blur matrix

Platform	Backend	Result
Windows 11	DWM acrylic / Mica via DirectComposition	True system blur; GPU cost lowest
Windows 10	Legacy acrylic; falls back to translucency	Blur on most GPUs, else tier 2
KDE Plasma 5.24+	org.kde.kwin.blur protocol	True backdrop blur, saturation supported
GNOME (Wayland / X11)	Aegis readback blur (GPU, cached)	Identical look without any OS protocol
X11, no compositor	Solid tint at 88%, hairline kept	Capsule remains fully legible
Reduce transparency	Tint 94%, blur off, borders strengthened	All silhouettes preserved

The glassbar must never render as a mystery rectangle: every tier keeps the capsule silhouette, hairline and trust pill.

Table 10 — Backdrop fallback ladder

Condition	Strategy
GPU budget healthy (default)	Live backdrop blur — OS acrylic where available, readback path elsewhere
Battery saver / low-end GPU	Cached quarter-res snapshot, refreshed on scroll-idle (200 ms throttle)
Blur unavailable (no compositor)	Solid tint at 88% opacity, hairline border kept
Accessibility "reduce transparency"	Tint 94%, blur off, borders strengthened; reduced motion collapses fades to 80 ms

One recipe, four tiers — the user never sees a broken state, only a calmer one.

Table 11 — Keyboard and screen-reader accessibility

Item	Requirement
Keyboard reach	Every chrome action reachable without pointer; full shortcut map in Table 3
Focus ring	2 dp ring, 2 dp offset, accent (invert) color; F6 cycles page → glassbar → sheet
Rail	Announced as "Show browser controls" toggle; 32 × 192 hit area
Sheets	Role dialog with focus trap; Esc returns focus to the capsule control that opened it
Contrast	Pill and tab text 4.5:1 on glass in both themes; tint tuned to pass over any page
Motion & transparency	Both OS settings honored — 80 ms fades, 94% tint (Tables 9-10)

Glass never carries meaning without text or glyph; states are always duplicated in text for screen readers.

Table 12 — Z-layer stack

z	Layer	Contents
0	WebContents	Live page, edge to edge, under the top band
1	Page UI	Find-in-page bar, HTML dialogs (page-owned)
2	Glassbar / rail	Liquid glass chrome, elevation 0–1
3	Sheets + scrim	Tab grid, menu, command omnibox; dim + blur behind
4	OS surfaces	IME windows, context menus, system toasts, snap previews

Layer 4 always wins — IME and system UI are never covered by Aegis chrome.

6.1 Ship sequence at a glance

Stage	Scope	Done when
M1 · Capsule	Frameless window, caption cluster, glass capsule with bound trust line, + and ⋮	Capsule renders over any page at 60 fps on Win11 and KDE; omnibox round-trips
M2 · States	Hover expansion, auto-hide triggers, edge rail with recall	Reading a long article shows only the rail; hover expansion measures ≤ 240 ms
M3 · Sheets	Tab grid, menu, command omnibox on the shared sheet component	All three open and dismiss under 400 ms with the live page blurred behind
M4 · Parity	Light theme, GNOME readback path, Win10 and reduce-transparency tiers	Token swap only; QA passes on Windows 10/11 and GNOME/KDE, both themes

Each milestone ships on its own; none of them requires compositor-level changes.

A pragmatic build order falls out of this stack: (1) the frameless window with the bound capsule, the smallest slice that already feels like Aegis on desktop; (2) the state machine — hover strip, auto-hide and the rail — which unlocks the full-window reading story; (3) the three sheets on the shared component; (4) platform parity, which is mostly token work plus the GNOME readback path. Each stage is independently shippable and none touches the compositor: the entire proposal lives in the application layer, which is exactly where a hardened browser should keep its interface decisions.